

Line Tracking and Obstacle Avoidance Robot



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Introduction

In modern engineering applications, such as industrial automation, navigation, and intelligent transportation systems, autonomous mobile robots are crucial. The design and execution of an autonomous mobile robot with tunnel detection, wall and obstacle avoidance, line following, and parking are presented in this project. The system combines one microcontroller-based control unit PIC16F877A with several sensors to provide dependable navigation and real-time decision making in a controlled setting. The robot successfully completes all necessary difficulty zones on its own by combining sensor fusion, motor control, and modular software design.

Design

The robot is built on a four-wheeled chassis driven by two pairs of independent DC motors, enabling precise control of the left and right sides. Three downward-facing IR sensors are used for line following, where the middle sensor maintains straight motion and the side sensors correct deviations from the black line. Two additional IR sensors are used for wall detection to prevent collisions during navigation, while an ultrasonic sensor measures distance for obstacle avoidance and parking.

During the parking stage, the robot detects the black parking zone using the three bottom IR sensors and activates a periodic buzzer. Once the ultrasonic sensor detects the parking wall, the robot performs a 180-degree right turn, stops all motors, turns off the LED, and raises a flag using a servo motor to indicate successful completion.

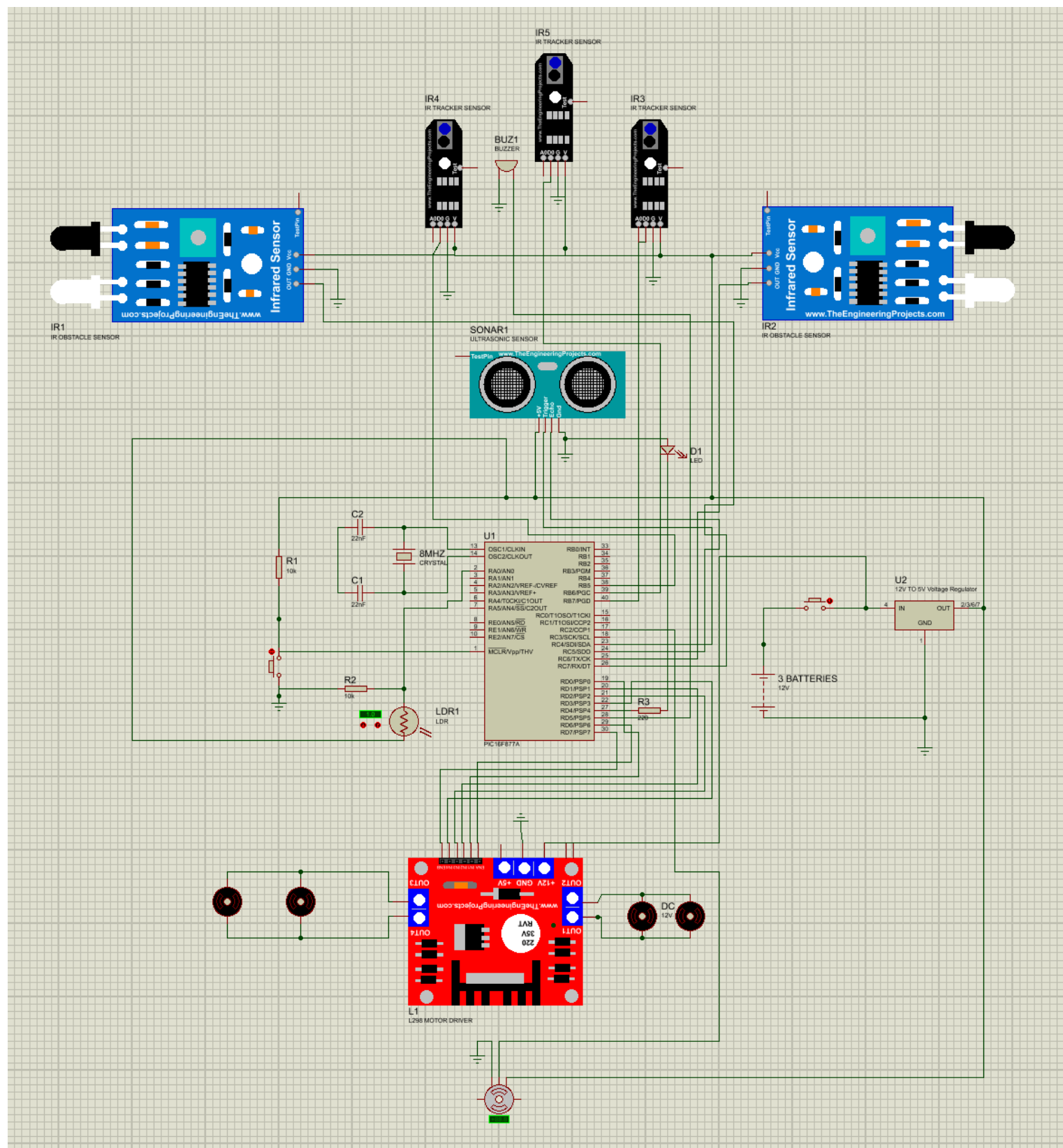


Figure 1: Hardware Design.

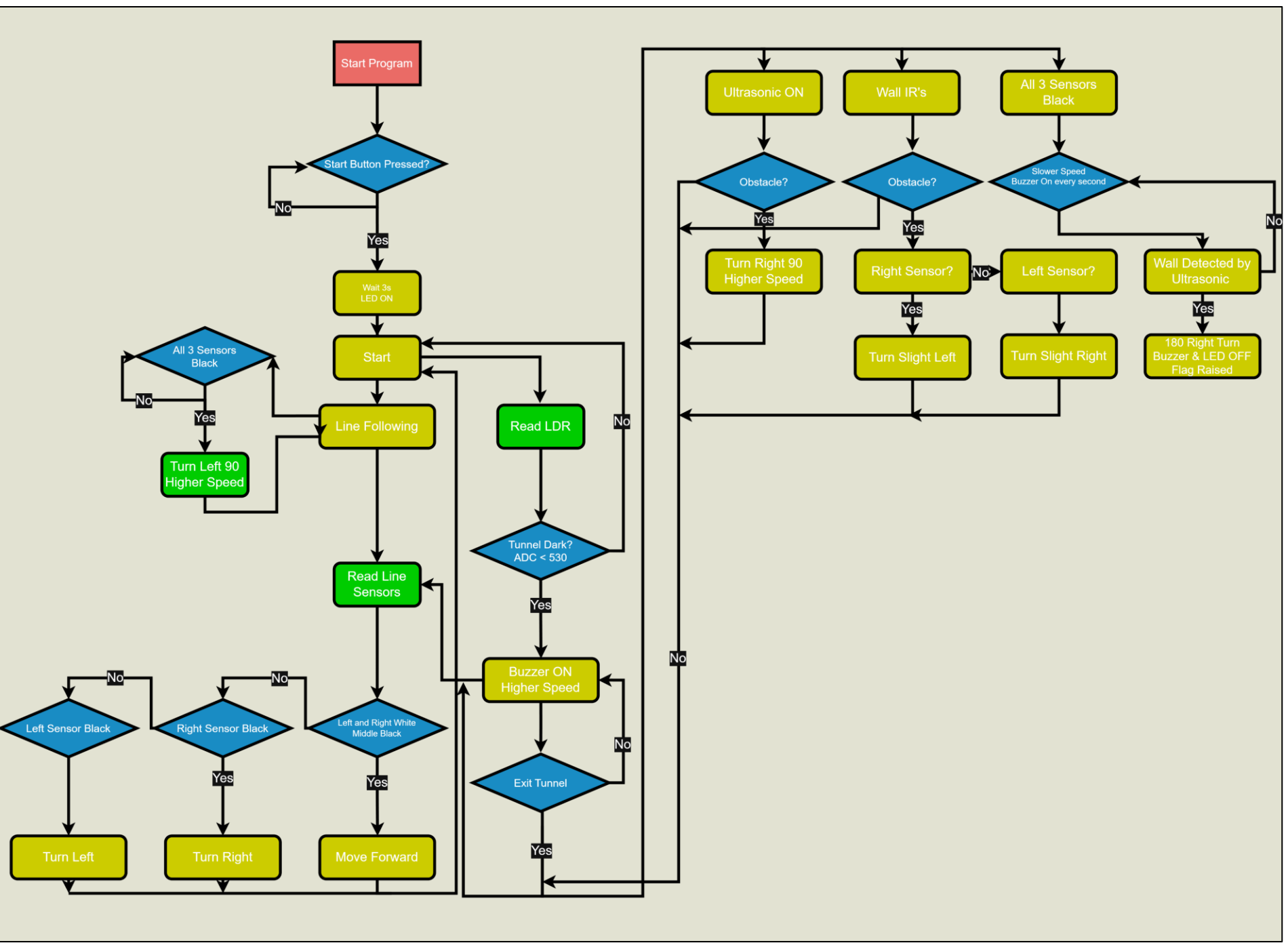


Figure 2: Software Design.

Results

The autonomous mobile robot successfully completed all required challenge zones without human intervention. The line-following algorithm demonstrated stable tracking under varying surface conditions, while the LDR-based tunnel detection reliably identified low-light environments and activated the buzzer. Ultrasonic sensing enabled effective obstacle detection and avoidance, with side IR sensors providing smooth wall-avoidance behavior. In the final stage, the robot accurately detected the parking zone, executed a controlled approach using distance feedback, performed a 180-degree right turn, and raised the flag using servo actuation. Overall, the system demonstrated reliable sensor integration and consistent real-time performance.

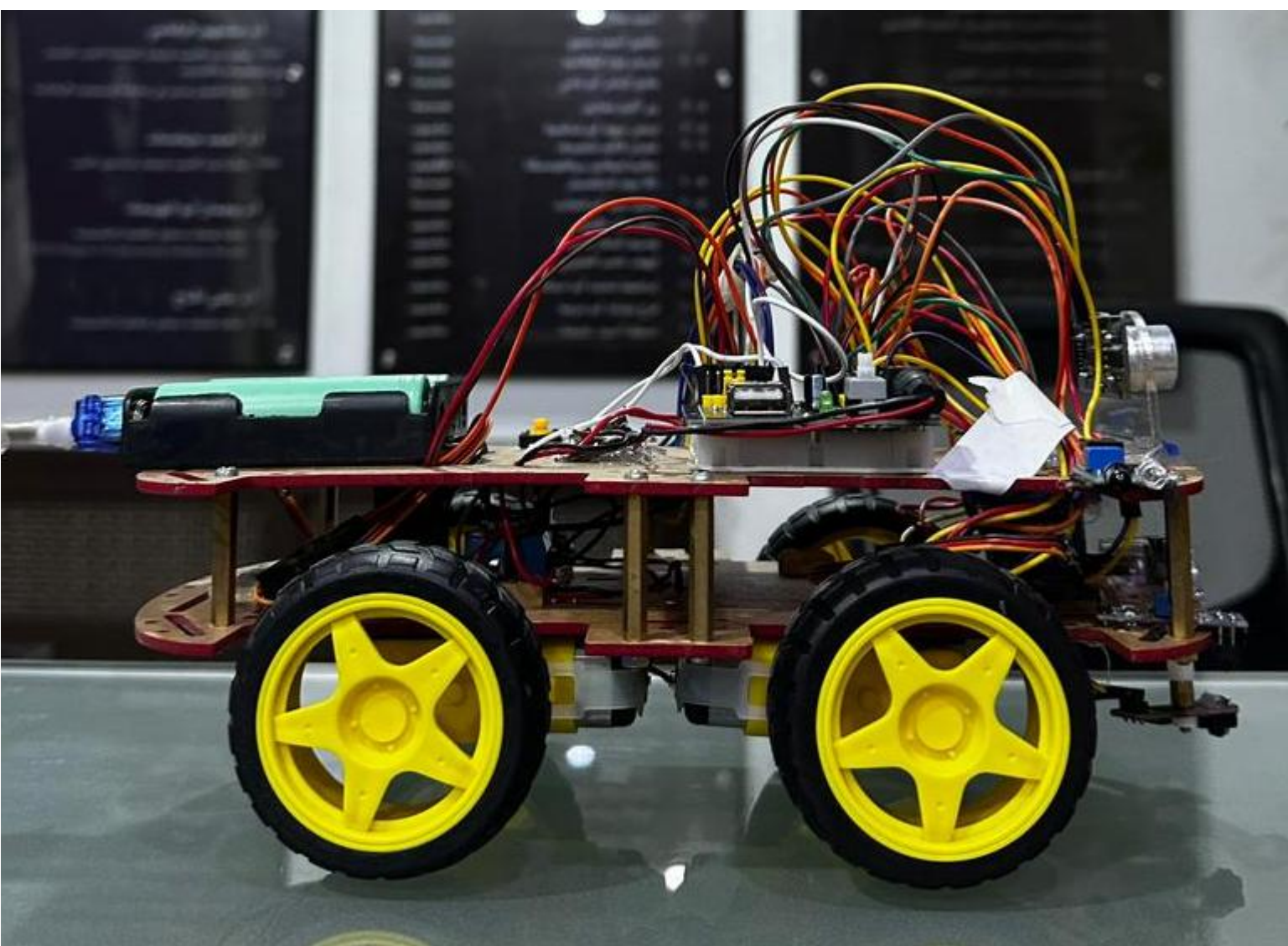


Figure 3: Side View

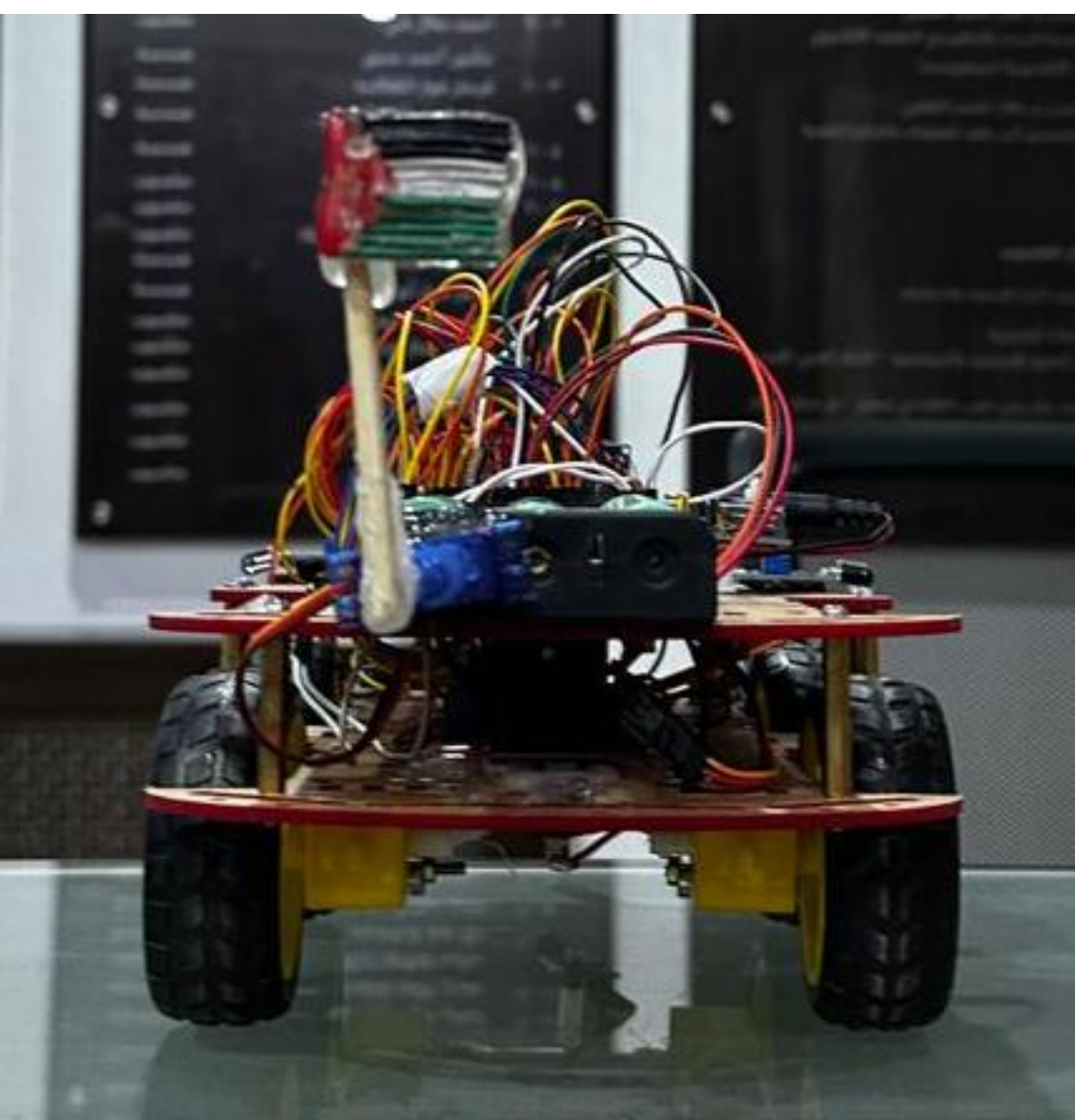


Figure 4: Back View

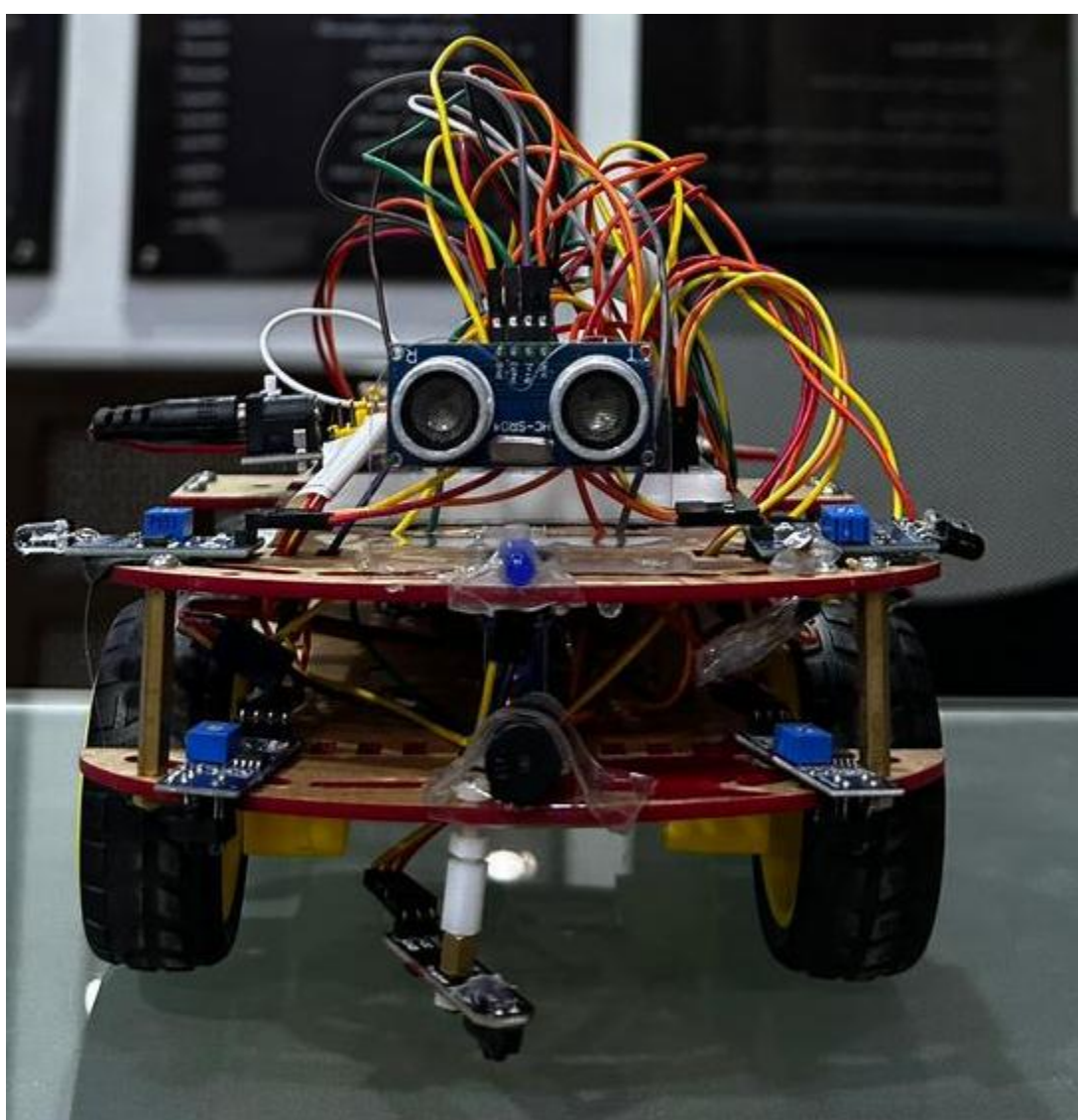


Figure 5: Front View

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Conclusion

This project demonstrates how a low-cost, sensor-based autonomous robot can reliably navigate complex environments without external control. By integrating multiple sensors with a structured control strategy, the robot successfully performs line following, obstacle and wall avoidance, tunnel detection, and precision parking. The significance of this work lies in showing that robust autonomous behavior can be achieved through efficient sensor fusion and modular design, making the approach applicable to real-world robotic systems such as automated vehicles, warehouse robots, and intelligent service platforms.